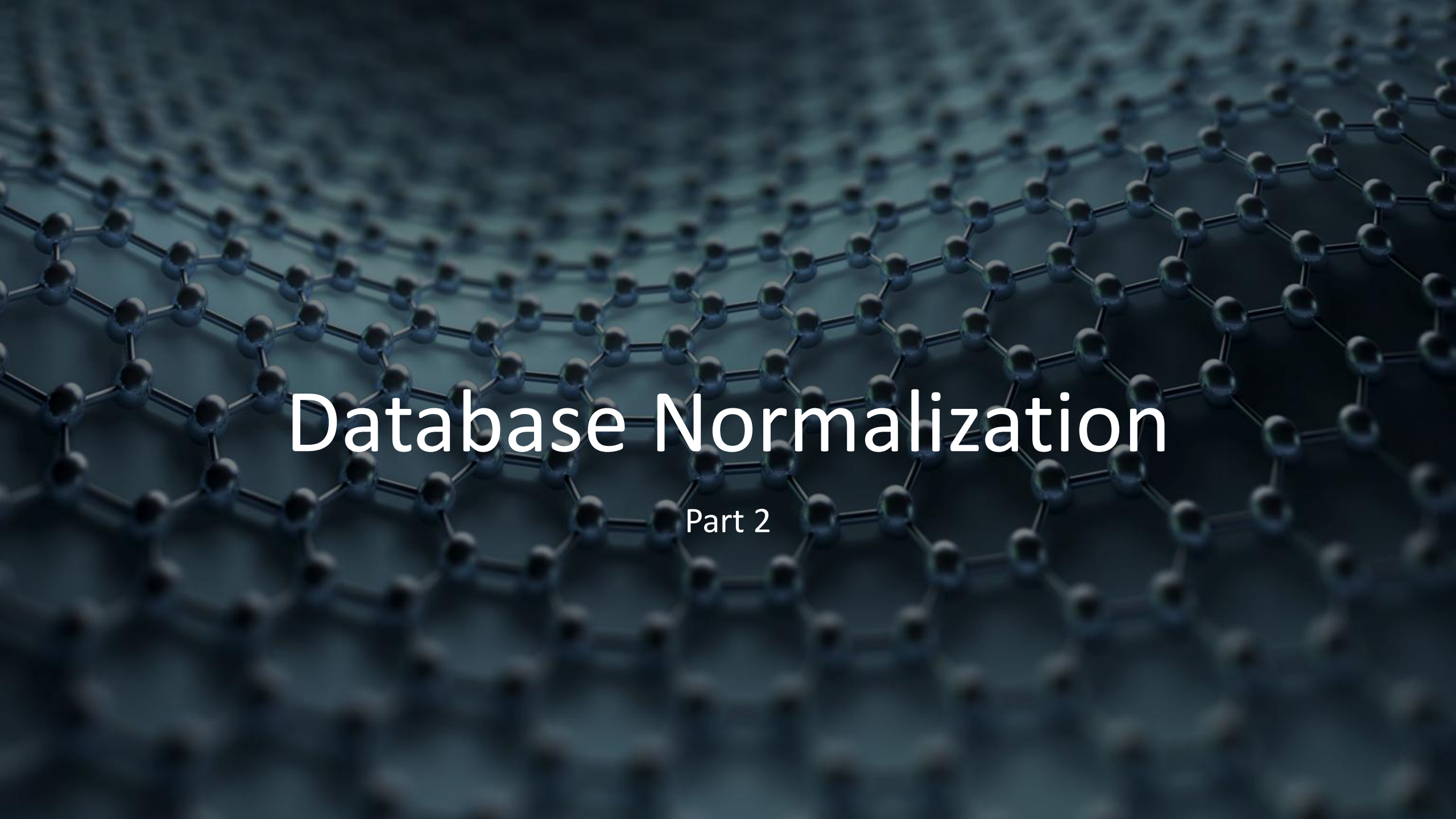




Database Normalization

Part 2

Objectives

- Demonstrate how 3rd Normal Form works
- Define what is transitive dependency.
- Present ways on how to eliminate transitive dependency.
- Differentiate Transitive Dependency from Partial Dependency
- Demonstrate how to transform a relational table to 3rd Normal Form.

What is Transitive Dependency?

- If the value of a non-primary attribute can be defined using another non-primary attribute, then it is called a transitive dependency.
- When any attribute does not require a primary key and can easily get value using another non-primary attribute, then it is called a **Transitive Dependency**.

Example:

roll_no	name	city	zip-code
1	abc	pune	411044
2	jkl	mumbai	400001
3	uvw	pune	411044
4	xyz	delhi	110001

What is Third Normal Form?

- What is the Third Normal Form in DBMS? A given relation is said to be in its third normal form when it's in 2NF but has no transitive partial dependency.

Conditions for Third Normal Form

- 1. Table must be in Second Normal Form
- 2. Table must not contain transitive dependency.

Hence,

- If any table satisfies these two conditions, then the table is in third normal form.



CONDITION 1: TABLE MUST BE IN SECOND NORMAL FORM

Emp_Id	Emp_Name	Mob_No
1	ABC	7485961232
2	DEF	8596742123
3	GHI	9674851323

FUNCTIONAL DEPENDENCIES :

EMP_ID → **EMP_NAME, MOB_NO**

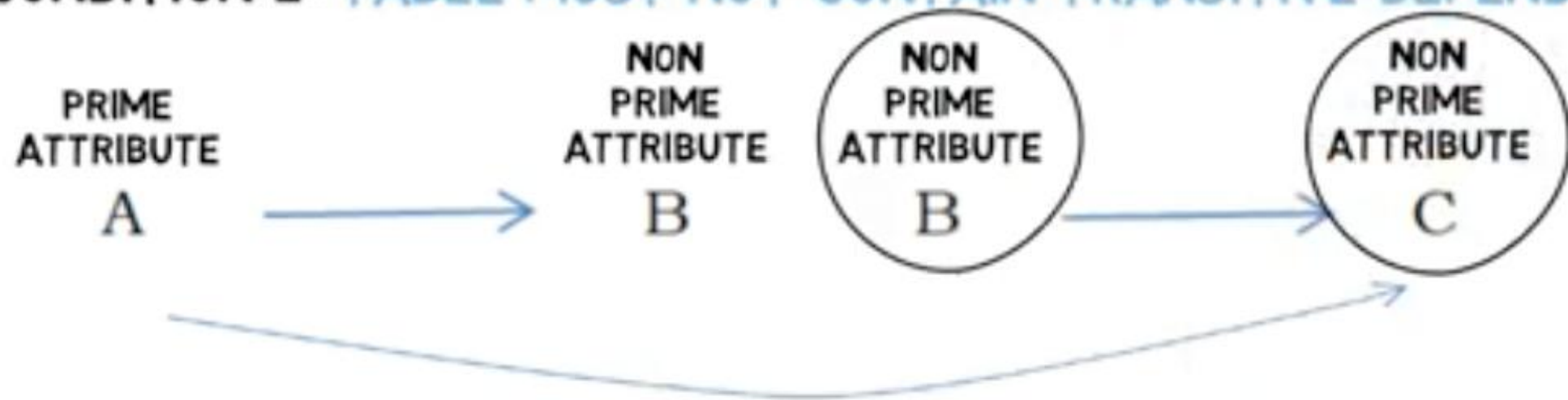
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NO COMPOSITE ATTRIBUTE NO MULTI-VALUED ATTRIBUTE NO PARTIAL DEPENDENCY

TABLE IS IN 1NF

TABLE IS IN 2NF

CONDITION 2: TABLE MUST NOT CONTAIN TRANSITIVE DEPENDENCY



Emp_ID $\longrightarrow$ Project_ID Project_ID $\longrightarrow$ Project_Name

PRIMARY KEY

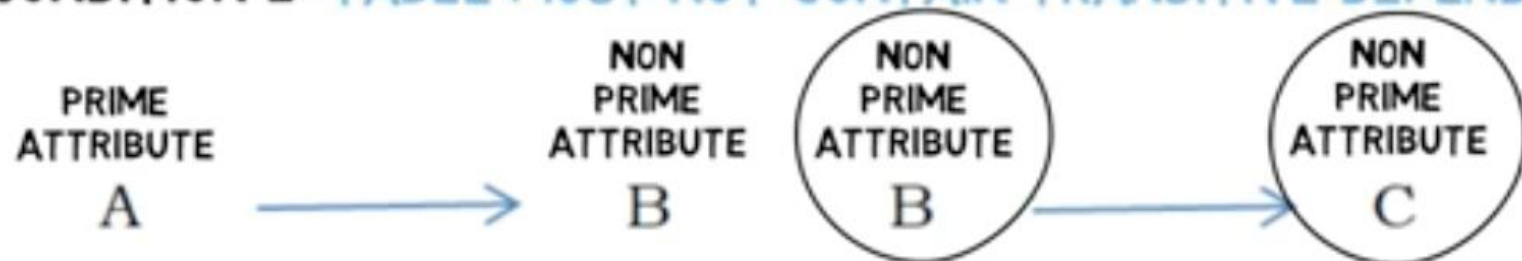
Emp_ID	Emp_Name	Project_ID	Project_Name
1	ABC	123	X
2	DEF	789	Z
3	GHI	123	X
4	JKL	123	X
5	MNO	789	Z
6	PQR	789	Z
7	STU	123	X

FUNCTIONAL DEPENDENCIES :

EMP_ID $\longrightarrow$ EMP_NAME, PROJECT_ID

PROJECT_ID $\longrightarrow$ PROJECT_NAME

CONDITION 2: TABLE MUST NOT CONTAIN TRANSITIVE DEPENDENCY



Emp ID → Project_ID → Project_Name

PRIMARY KEY

Emp_ID	Emp_Name	Project_ID	Project_Name
1	ABC	123	X
2	DEF	789	Z
3	GHI	123	X
4	JKL	123	X
5	MNO	789	Z
6	PQR	789	Z
7	STU	123	X

FUNCTIONAL DEPENDENCIES :

EMP_ID → EMP_NAME, PROJECT_ID

PROJECT_ID → PROJECT_NAME

Solution

Emp_ID	Emp_Name	Project_ID	Project_Name
1	ABC	123	X
2	DEF	789	Z
3	GHI	123	X
4	JKL	123	X
5	MNO	789	Z
6	PQR	789	Z
7	STU	123	X

Primary Key Foreign Key

Emp_ID	Emp_Name	Project_ID
1	ABC	123
2	DEF	789
3	GHI	123
4	JKL	123
5	MNO	789
6	PQR	789
7	STU	123

Functional Dependencies:

Emp_ID $\longrightarrow$ Emp_Name, Project_ID

Primary Key

Project_ID	Project_Name
123	X
789	Z

Functional Dependencies:

Project_ID $\longrightarrow$ Project_Name

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Partial vs. Transitive Dependency

CONDITION 2: THERE MUST NOT BE ANY PARTIAL DEPENDENCY

Student_ID	Course_ID	Course_Fee
1	IOT	750
2	IOT	750
3	IOT	750
1	AI	880
2	AI	880
3	AI	880

(Student_ID, Course_ID) → Course_Fee
COMPLETE CANDIDATE KEY NON-PRIME ATTRIBUTE

PARTIAL DEPENDENCY
Course_ID → Course_Fee
PART OF CANDIDATE KEY NON-PRIME ATTRIBUTE

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Prime Attributes	Non-Prime Attributes
Student_ID	Course_Fee
Course_ID	

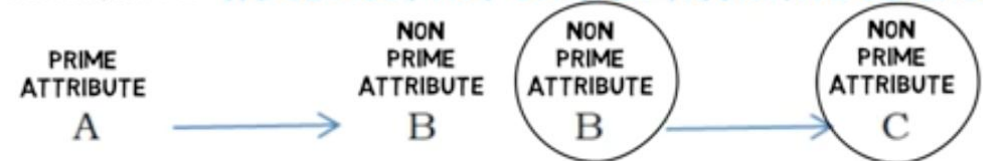


(Student_ID, Course_ID)
IS A CANDIDATE KEY

IF A NON-PRIME ATTRIBUTE IS DETERMINED BY A PART OF CANDIDATE KEY, THAT FUNCTIONAL DEPENDENCY IS PARTIAL DEPENDENCY

Partial Dependency: A non-prime attribute is dependent on part of a composite primary key.

CONDITION 2: TABLE MUST NOT CONTAIN TRANSITIVE DEPENDENCY



Emp_ID → Project_ID → Project_Name

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PRIMARY KEY

Emp_ID	Emp_Name	Project_ID	Project_Name
1	ABC	123	X
2	DEF	789	Z
3	GHI	123	X
4	JKL	123	X
5	MNO	789	Z
6	PQR	789	Z
7	STU	123	X

FUNCTIONAL DEPENDENCIES:

EMP_ID → EMP_NAME, PROJECT_ID

PROJECT_ID → PROJECT_NAME

Transitive Dependency: A non-prime attribute depends on another non-prime attribute.

Why 3rd Normal Form?

- We use the 3NF to reduce any duplication of data and achieve data integrity in a database.
- The third normal form is fit for the designing of normal relational databases.
- It is because a majority of the 3NF tables are free from the anomalies of deletion, updates, and insertion.

Activity 1: Convert the table into 3rd Normal Form

OrderID	CustomerName	CustomerAddress	ProductID	ProductName	Quantity	UnitPrice	TotalPrice	OrderDate	CustomerPhone
1	John Smith	123 Main St	101	Laptop	2	500	1000	2023-09-15	555-1234
2	Jane Doe	456 Oak St	102	Mouse	1	20	20	2023-09-16	555-5678
3	John Smith	123 Main St	103	Keyboard	1	30	30	2023-09-17	555-1234
4	John Smith	123 Main St	102	Mouse	2	20	40	2023-09-17	555-1234
5	Jane Doe	456 Oak St	101	Laptop	1	500	500	2023-09-18	555-5678

Activity 1: Table in 3rd Normal Form (Answer)

Table: Customer (Stores customer information)

<u>CustomerID</u>	Last Name	First Name	<u>CustomerAddress</u>	<u>CustomerPhone</u>
001	Smith	John	123 Main St	555-1234
002	Doe	Jane	456 Oak St	555-5678

Table: Products (Stores product-related information)

<u>ProductID</u>	ProductName	<u>UnitPrice</u>
101	Laptop	500
102	Mouse	20
103	Keyboard	30

Table: Order (Tracks when the order was made, links customer and product)

Order ID	<u>CustomerID</u>	<u>OrderDate</u>
1	1	2023-09-15
2	2	2023-09-16
3	1	2023-09-17
4	1	2023-09-17
5	2	2023-09-18

Table: Order Details (Product and Quantity information for each order)

Order ID	<u>ProductID</u>	Quantity	<u>TotalPrice</u>
1	101	2	1000
2	102	1	20
3	103	1	30
4	102	2	40
5	101	1	500

Activity 2: Convert to 3rd Normal Form

AssignmentID	EmployeeID	EmployeeName	ProjectID	ProjectName	HoursWorked
1	E101	John Smith	P201	Website	40
2	E102	Jane Doe	P202	Mobile App	35
3	E101	John Smith	P202	Mobile App	20
4	E103	Mark Twain	P201	Website	50

Activity 2: Convert to 3rd Normal Form

AssignmentID	EmployeeID	EmployeeName	ProjectID	ProjectName	HoursWorked
1	E101	John Smith	P201	Website	40
2	E102	Jane Doe	P202	Mobile App	35
3	E101	John Smith	P202	Mobile App	20
4	E103	Mark Twain	P201	Website	50

Business Rule Observations:

- An employee can be assigned many projects.
- A project can be assigned to many employees.

Activity 2: Convert to 3rd Normal Form (Answer)

Table: Employee

EmployeeID	EmployeeName
E101	John Smith
E102	Jane Doe
E103	Mark Twain

Table: Project

ProjectID	ProjectName
P201	Website
P202	MobileApp

Table: Assignment

AssignmentID (PK)	EmployeeID (FK)	ProjectID(FK)	HoursWorked
1	E101	P201	40
2	E102	P202	35
3	E101	P202	20
4	E103	P201	50

Activity 3: Convert to 3rd Normal Form

EmployeeID	EmployeeName	EmployeeEmail	DepartmentID	DepartmentName	ProjectID	ProjectName	ProjectStartDate	ProjectEndDate	HoursAssigned
E001	Alice Brown	alice@example.com	D001	HR	P001	Recruitment	2023-07-01	2023-10-01	20
E002	Bob White	bob@example.com	D002	IT	P002	Cloud Migration	2023-08-01	2023-12-01	30
E001	Alice Brown	alice@example.com	D001	HR	P003	Training	2023-09-01	2023-11-01	10
E003	Charlie Green	charlie@example.com	D002	IT	P002	Cloud Migration	2023-08-01	2023-12-01	40
E002	Bob White	bob@example.com	D002	IT	P004	Security Audit	2023-09-15	2023-12-15	25

Activity 3: Convert to 3rd Normal Form (Answer)

Table: Employee

EmployeeID	EmployeeName	EmployeeEmail
E001	Alice Brown	alice@example.com
E002	Bob White	bob@example.com
E003	Charlie Green	charlie@example.com

Table: Department

DepartmentID	DepartmentName
D001	HR
D002	IT

Table: Project

ProjectID	ProjectName	ProjectStartDate	ProjectEndDate
P001	Recruitment	2023-07-01	2023-10-01
P002	Cloud Migration	2023-08-01	2023-12-01
P003	Training	2023-09-01	2023-12-01
P004	Security Audit	2023-09-15	2023-12-15

Table: Assignment (Junction/Associative/Bridge Table)

EmployeeID (FK)	ProjectID (FK)	HoursAssigned
E001	P001	20
E002	P002	30
E001	P003	10
E003	P002	40
E002	P004	25

Answers

Why Data Normalization?

- Improving the logical design to create well-structured relations to:
 1. Avoid duplication and conserve storage.
 2. Satisfy certain referential integrity constraints (partial row in one table can be related to at most one row in a related table)
 3. Facilitate data maintenance by eliminating the insertion, update and deletion anomalies.
 4. Provide better design that enables future growth

Normalize the given table into 3rd Normal Form

Employee_ID	Employee_Name	Department	Department_Location	Project_ID	Project_Name	Hours_Worked	Pay_Rate	Gross_Pay	Tax_Deduction	Insurance_Deduction	Net_Pay
1001	Alice Johnson	HR	New York	P001	HR Software	40	50	2000	200	100	1700
1002	Bob Smith	IT	Chicago	P002	IT Infrastructure	38	60	2280	228	150	1902
1001	Alice Johnson	HR	New York	P003	Employee Training	10	50	500	50	30	420
1003	Charlie Lee	Finance	San Francisco	P004	Financial Audit	45	55	2475	247	120	2108
1002	Bob Smith	IT	Chicago	P003	Employee Training	15	60	900	90	70	740